

LECTURE NOTE ON THE DATA PROCESSING, ANALYSIS AND INTERPRETATION (May, 2019)

INTRODUCTION

The data, after collection, has to be processed and analyzed in accordance with the outline laid down for the purpose at the time of developing the research proposal. Data processing involves editing, coding, classification and tabulation of the data collected so that they are amenable to analysis. This is an intermediary stage between the collection of data and their analysis and interpretation.

6.1: DATA PROCESSING OPERATIONS

Technically speaking, processing implies editing, coding, classification and tabulation of collected data so that they are amenable to analysis.

1) ***Editing:*** Editing is a process of examining the collected raw data to detect errors and omissions and to correct these when possible. Editing is done to assure that the data are accurate, consistent with other facts gathered, uniformly entered, as complete as possible and have been well arranged to facilitate coding and tabulation. With regard to points or stages at which editing should be done, one can talk of field editing and central editing.

Field editing consists in the review of forms by the investigator for completing what the interviewer (enumerator) has written in abbreviated and/or in illegible form at the time of recording the respondents' responses. This type of editing should be done as soon as possible after the interview, preferably on the very day or on the next day. While doing field editing, the investigator must restrain himself and must not correct errors of omission by simply guessing what the informant would have said if the question had been asked. Preliminary editing by a field supervisor on the interview to catch technical omissions, check legibility of hand writing and clarify responses that are logically inconsistent

Central editing/in house editing should take place when all forms of questionnaires have been completed and returned to the office. Editor(s) may correct the obvious errors such as an entry in the wrong place, entry recorded in months when it should have been recorded in weeks, and the like. At times the respondents can be asked for clarification. The editor must strike out the answer

if the same is inappropriate and he has no basis for determining the correct answer or the response. In such a case an editing entry of 'no answer' is called for.

2) **Coding:** Coding refers to the process of assigning numerals or other symbols to answers so that responses can be put into a limited number of categories or classes. These classes must possess the characteristic of exhaustiveness (there must be a class for every data item) and also that of mutual exclusivity which means that a specific answer can be placed in one and only one cell in a given category set.

Coding is necessary for efficient analysis and through it the several replies may be reduced to a small number of classes which contain the critical information required for analysis. Coding decisions should usually be taken at the designing stage of the questionnaire. In case of hand coding, it is possible to code on the margin of the questionnaire with colored pencil or to transcribe the data from the questionnaire to a coding sheet.

3) **Classification:** Classification is the process of arranging data in groups or classes on the basis of common characteristics, especially for studies with large volume of raw data. Data having a common characteristic are placed in one class and in this way the entire data get divided into a number of groups or classes. Classification can be according to attributes based on descriptive information (such as literacy, sex, honesty, income level, prestige, educational level etc.). A researcher, based on attributes, can classify in to two classes, one class consisting of items possessing the given attribute and the other class consisting of items which do not possess the given attribute.

For instance, if an investigator has gathered information on the occupation of several hundred individuals with the following occupational lists (Lawyers, Barbers, Practical nurse, Carpenters, Brokers, Elevator operator, Veterinarian, Migrant farm laborer, Executive, Engineer, Advertising agent, Electrician)

Before this data can be analyzed the researcher need to organize the occupations in to different categories. The following is one possible way of classifying occupations into categories.

- Professional and Managerial: Layers, Veterinarians, Executive, Engineers
- Service and Skilled Labors: Barbers, Elevator operator, Practical nurse, Electrician, carpenter

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- Professional Technical and Sales: Advertising agent, broker ; and
- Unskilled Labor: Migrant farm laborers

This system classifies occupations according to the level of income, prestige, and the education that they have in common, and permit the researcher to use four well defined categories in his/her analysis rather than dozens of specific occupations. Such classification is known as coding scheme.

Classification can also be according to class-intervals. Numerical characteristics relating to income, production, age, weight, etc. can be classified based on class-intervals. For instance, persons whose incomes, say, are within Br 201 to Br 400 can form one group; those whose incomes are within Br 401 to Br 600 can form another group and so on. The difference between the two class limits is known as class magnitude. The number of items which fall in a given class is known as the frequency of the given class.

4) Tabulation: When a mass of data has been assembled, it becomes necessary for researcher to arrange the same in some kind of concise and logical order. This procedure is referred to as tabulation. Thus tabulation is the process of summarizing raw data and displaying the same in compact form for further analysis. Tabulation is essential because of the following reasons:

- i) It conserves space and reduces explanatory and descriptive statement to a minimum
- ii) It facilitates the process of comparison
- iii) It facilitates the summation of items and the detection of errors and omissions
- iv) It provides a basis for various statistical computations.

6.2: Measurement Scales

Before analyzing data, it is important to identify the measurement scales of the data type. Measurement is defined as: The determination of size in relation to some observed standard, e.g. metre, kilogram, second, ampere, degree Kelvin, candela, mole, or some unit derived from these seven basic units. Measurement is the process observing and recording the observations. Measurement is the estimation or determination of extent, dimension or capacity, usually in relation to some standard or unit of measurement.

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The measurement is expressed as a number of units of the standard, such as distance being indicated by a number of kilometers. According to S.S.Stevens, Measurement is the assignment of numerals to objects or events according to rules. According to Campbell, “Measurement is the assignment of numbers to represent properties.

There are four basic measurement scales: nominal, ordinal, interval, and ratio. The most accepted basis for scaling has three characteristics:

- ✓ Numbers are ordered. One number is less than, greater than, or equal to another number.
- ✓ Differences between numbers are ordered. The difference between any pair of numbers is greater than, less than, or equal to the difference between any other pair of numbers.
- ✓ The number series has a unique origin indicated by the number zero.

Combination of these characteristics of order, distance, and origin provide the following widely used classification of measurement scales.

Nominal Scales: When we use nominal scale, we partition a set into categories that are mutually exclusive and collectively exhaustive. The counting of members is the only possible arithmetic operation and as a result the researcher is restricted to the use of the mode as the measure of central tendency. If we use numbers to identify categories, they are recognized as labels only and have no quantitative value. Nominal scales are the least powerful of the four types. They suggest no order or distance relationship and have no arithmetic origin. Examples can be respondents' marital status, gender, students' Id number, etc.

Ordinal Scales: Ordinal scales include the characteristics of the nominal scale plus an indicator of order. The use of an ordinal scale implies a statement of 'greater than' or 'less than' (an equality statement is also acceptable) without stating how much greater or less. Thus the real difference between ranks 1 and 2 may be more or less than the difference between ranks 2 and 3. The appropriate measure of central tendency for ordinal scales is the median. Examples of ordinal scales include opinion or preference scales.

Interval Scales: The interval scale has the powers of nominal and ordinal scales plus one additional strength. It incorporates the concept of equality of interval (the distance between 1 and 2 equals the distance between 2 and 3). When a scale is interval, you use the arithmetic mean as the measure of central tendency. Calendar time is such a scale. For example, the elapsed time

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between 4 and 6 A.M. equals the time between 5 and 7 A.M. One cannot say, however, 6 A.M is twice as late as 3 A.M. because zero time is an arbitrary origin. Centigrade and Fahrenheit temperature scales are other examples of classical interval scales.

Ratio Scales: Ratio scales incorporate all of the powers of the previous ones plus the provision for absolute zero or origin. The ratio scale represents the actual amounts of a variable. Multiplication and division can be used with this scale but not with the other mentioned. Money values, population counts, distances, return rates, weight, height, and area can be examples for ratio scales. Ratio measurement has a true zero.

A true zero means that the object measuring zero possesses none of the property in question. Height and weight are obvious example. In general, wherever the objects, are being counted‘ for possessing or not possessing a certain characteristic ratio measurement is being done. In all such cases the number 0 has a true meaning – the object possesses none of the characteristic being measured. Other common examples of this type of measurement are sales, costs, number of purchasers, length, time, etc.

6.3: DATA ANALYSIS

The term analysis refers to the computation of certain measures along with searching for patterns of relationship that exist among data-groups. Analysis involves estimating the values of unknown parameters of the population and testing of hypotheses for drawing inferences. Analysis can be categorized as descriptive analysis and inferential (statistical) analysis.

6.3.1: Descriptive Analysis

With descriptive statistics, there is no uncertainty because you are describing only the people or items that you actually measure. You’re not trying to infer properties about a larger population.

It is largely the study of distribution of one variable. The characteristics of location, spread, and shape describe distributions. Their definitions, applications, and formulas fall under the heading of descriptive statistics. They describe and summarize data using a few indices or statistics. They include frequency distributions, measures of central tendencies (mean, median, and mode); measures of variability/spread/ or dispersion (range, variance, and standard deviations); and common measures of shape are *skewness and kurtosis*. Skewness is a measure of symmetry, or more precisely, the lack of symmetry. A distribution, or data set, is symmetric if it looks the same

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to the left and right of the center point. Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers. A uniform distribution would be the extreme case. Kurtosis refers to the pointedness of a peak in the distribution curve. Frequency Distribution- it shows the distribution (or the count) of individual scores in a sample for a specific variables. A frequency table indicates “how many times” each score or response occurs.

6.3.2: Inferential Analysis

It includes two topics, estimation of population values and testing statistical hypothesis. We may as well talk of correlation analysis and causal analysis. Correlation analysis studies the joint variation of two or more variables for determining the amount of correlation between two or more variables. Causal (regression) analysis is concerned with the study of how one or more variables affect changes in another variable. It is thus a study of functional relationships existing between two or more variables. It is about using data from sample and then making inferences about the larger population from which the sample is drawn. The goal of the inferential statistics is to draw conclusions from a sample and generalize them to the population. It determines the probability of the characteristics of the sample using probability theory.

In inferential statistics, a primary goal is to estimate population parameters. These parameters are the unknown values for the entire population, such as the population mean and standard deviation. These parameter values are not only unknown but almost always unknowable. Typically, it's impossible to measure an entire population. The sampling error I mentioned earlier produces uncertainty, or a margin of error, around our estimates.

6.3.2.1: CORRELATION ANALYSIS

In case of bivariate or multivariate populations, we often wish to know the relation of the two and/or more variables in the data to one another. We may like to know, for example, whether the number of hours workers devote for leisure is somewhat related to their income, to age, to sex, to education level or to similar other factors. We may ask ‘Is there any association or correlation between the two (or more) variables? If yes, of what degree?’ These questions are answered by the use of correlation analysis. Various forms of correlation coefficients can be computed in indices that measure relationship or association, some of which are presented below

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If the correlation between two variables exists or not, we have to find out

- i) Whether or not the correlation ship exists or not ii) Degree of correlation;
- iii) Is it positive or negativities, i.e. direct or indirect, and
- iv) Is the relationship strong or weak This requires determining coefficient of correlation. Simple correlation coefficient is the most popular statistical measure that indicated the relationship between the two variables.

The correlation coefficient ranges from +1 to -1. If the value is +1, there is a perfect positive relationship, but if the value is -1, a perfect negative relationship or a perfect inverse relationship is indicated. If the value is zero, no relationship is indicated.

Correlation is a statistical technique used for measuring the relationship or interdependence of two or more variables. The measurement scales used should be at least interval scales, but other correlation coefficients are available to handle other types of data. Correlation coefficients can range from -1.00 to +1.00. The value of -1.00 represents a perfect negative correlation while a value of +1.00 represents a perfect positive correlation. A value of 0.00 represents a lack of correlation.

In statistics, the **Correlation coefficient** (r) is a measure of how well a linear equation describes the relation between two variables X and Y measured on the same object or organism.

6.3.2.2: REGRESSION ANALYSIS

Regression describes the relationship between a set of independent variables and a dependent variable. This analysis incorporates hypothesis tests that help determine whether the relationships observed in the sample data actually exist in the population. The statistical tool with the help us to estimate (or predict) the unknown values of one variable from known values of another variable is called regression. For example if we know that advertising and sales are correlated, we may find out the expected amount of sales for a given advertising expenditure or the required amount of expenditure for attaining a given amount of sales. If we take two variables x and y, we shall have two regression lines as under.

In regression analysis, there will be one dependent variable and one or more independent variables. If a researcher is predicting the value of one dependent variable based on the value of one independent variable, then that is known as simple regression analysis. On the other hand, if

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the researcher uses the value of many independent variables to predict the value of one dependent variable, then it is known as multiple regression analysis.

Simple Regression Analysis – In using simple regression to predict the value of dependent variable from the known value of independent variable, one should follow two steps.

- Determining linear regression line by using mathematical equation
- Predicting the value of dependent variable by using the mathematical equation

The linear regression line shows the relationship between two variables (the dependent and independent variables). The following formula is used when we want to predict the unknown value of dependent variable based on the known value of independent variable.

$$Y_i = a + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 \dots \beta_n$$

Regression analysis can predict the outcome of a given key business indicator (dependent variable) based on the interactions of other related business drivers (independent variable). For example, it allows predicting sales volume, using the amount spent on advertising and number of sales people that a company employs.

The coefficient of determination, r^2 , is useful because it gives the proportion of the variance (fluctuation) of one variable that is predictable from the other variable. It is a measure that allows us to determine how certain one can be in making predictions from a certain model/graph. The coefficient of determination is the ratio of the explained variation to the total variation. The coefficient of determination is such that $0 < r^2 < 1$, and denotes the strength of the linear association between x and y.

The coefficient of determination represents the percent of the data that is the closest to the line of best fit. In regression analysis, the coefficient of determination is a measure of goodness-of-fit (i.e. how well or tightly the data fit the estimated model).

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6.3. Hypothesis testing

A **statistical hypothesis** is an assumption about a population parameter. This assumption may or may not be true.

There are two types of statistical hypothesis.

- **Null hypothesis.** The null hypothesis is usually the hypothesis that sample observations result purely from chance effects.
- **Alternative hypothesis.** The alternative hypothesis is the hypothesis that sample observations are influenced by some non-random cause.
- The term –null means nothing or invalid. It may be written as:

$$H_0: \mu = \mu_0$$

Where, H_0 is the null hypothesis and μ_0 is the mean of the population.

The alternative hypothesis is

$$H_A: \mu \neq \mu_0$$

The rejection of the null hypothesis will show that the mean of the population is not μ_0 . If this happens, then it implies that the alternative hypothesis is accepted. There can be more than two or more alternative hypothesis though only one can be tested at a time against the null hypothesis.

➤ **Hypothesis Testing**

Statisticians follow a formal process to determine whether to accept or reject a null hypothesis, based on sample data. This process, called **hypothesis testing**, consists of four steps.

- **Formulating the hypothesis:** The first step in hypothesis testing is to formulate the hypothesis to be tested. This means stating the null hypothesis and the alternative hypothesis.
- **Identifying the test statistic:** The test statistic is a statistic that will be used by the researcher to determine whether the null hypothesis should be accepted or rejected. Typically, the test statistic is the sample estimate of the population parameter in the null hypothesis. Therefore, since we are testing a hypothesis about a population mean, the test statistic will be the sample mean. When the hypothesis pertains to a large sample (30 or more), the z-test implying normal distribution is used. When a sample is small (less than 30), the use of the z-test will be inappropriate. Instead the t-test will be more

suitable. The test criteria frequently used in hypothesis testing are Z, t, F and χ^2 .

- **Formulating a decision rule:** The decision rule consists of two parts: (1) a test statistic and (2) a range of values, called the region of acceptance. The decision rule determines whether a null hypothesis is accepted or rejected. If the test statistic falls within the region of acceptance, the null hypothesis is accepted; otherwise, it is rejected.
- **Accepting or rejecting the null hypothesis:** Once the region of acceptance is defined, the null hypothesis can be tested against sample data. The test statistic is computed. For example consider that the test statistic is the sample mean. If the sample mean falls within the region of acceptance, the null hypothesis is accepted; if not, it is rejected.

Decision Rules

The decision rule is a procedure that a researcher uses to decide whether to accept or reject the null hypothesis. There are two types of errors that can result from a decision rule.

Type I error. A Type I error occurs when the researcher rejects a null hypothesis when it is true. The probability of committing a Type I error is called the **significance level**. This probability is also called **alpha**, and is often denoted by α .

Type II error. A Type II error occurs when the researcher accepts a null hypothesis that is false. The probability of committing a Type II error is called **Beta**, and is often denoted by β . The probability of *not* committing a Type II error is called the **Power** of the test.

In practice, the decision rule has two parts: (1) a test statistic and (2) a range of values. The range of values is called the **region of acceptance**. The region of acceptance is defined so that the chance of making a Type I error is equal to the significance level. If the test statistic falls within the region of acceptance, the null hypothesis is accepted.

Note: The set of values outside the region of acceptance is called the **region of rejection**. If the test statistic falls within the region of rejection, the null hypothesis is rejected. In such cases, we say that the hypothesis has been rejected at the α level of significance.

One-Tailed and Two-Tailed Tests

To understand the difference between these look at the table below. It shows three sets of hypothesis. Each makes a statement about how the population mean μ is related to a specified value M .

Set	Null hypothesis	Alternative hypothesis	Number of tails
1	$\mu = M$	$\mu \neq M$	2
2	$\mu \geq M$	$\mu < M$	1
3	$\mu \leq M$	$\mu > M$	1

The first set of hypotheses (Set 1) is an example of a two-tailed test, since an extreme value on either side of the sampling distribution would cause a researcher to reject the null hypothesis. The other two sets of hypotheses (Sets 2 and 3) are one-tailed tests, since an extreme value on only one side of the sampling distribution would cause a researcher to reject the null hypothesis.

A test of a statistical hypothesis, where the region of rejection is on only one side of the sampling distribution, is called a **one-tailed test**. For example, suppose the null hypothesis states that the mean is less than or equal to 10. The alternative hypothesis would be that the mean is greater than 10. The region of rejection would consist of a range of numbers located on the right side of sampling distribution; that is, a set of numbers greater than 10.

A test of a statistical hypothesis, where the region of rejection is on both sides of the sampling distribution, is called a **two-tailed test**. For example, suppose the null hypothesis states that the mean is equal to 10. The alternative hypothesis would be that the mean is less than 10 or greater than 10. The region of rejection would consist of a range of numbers located on both sides of sampling distribution; that is, the region of rejection would consist partly of numbers that were less than 10 and partly of numbers that were greater than 10.

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